

Simulation analysis of multi-ring interference assembly of large capacity composite flywheel rotor

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Abstract

The large-capacity power flywheel energy storage system serves as a high-quality frequency modulation resource for the power system. Utilizing high-strength, low-density composite materials in the manufacture of flywheel rotors is a primary method for enhancing flywheel energy storage. In this paper, we focus on the large-size multi-ring composite flywheel rotor. Based on the elastic theory, the stress distribution formula of the anisotropic material rotor rim under high-speed rotation is derived. Based on the stress superposition principle, the stress analysis formula under the interference fit of the composite rim and the metal hub is obtained, and the analytical solution is given. Based on the radial displacement of each ring, a suitable amount of interference is determined. Subsequently, a finite element analysis model for the interference fit between the composite rim and the metal hub is established. The stress distribution of the rotor is simulated and analyzed. The simulation results are basically consistent with the analytical results, which verifies the rationality of the model. Finally, we analyze and compare the difference between multi-ring isomorphism and multi-ring isomerism, and then the applicability of the analytical solution and simulation solution to the stress distribution of three-ring and four-ring composite flywheel rotor is further verified. The results demonstrate that, for large-size composite flywheels, existing formula analysis calculations and finite element simulation calculations align, highlighting a need for experimental verification in future research.

Keywords

Composite materials, flywheel rotor, analytical model, stress distribution, finite element analysis

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Introduction

Energy storage is a crucial technology that supports the new power system based on renewable energy, helping to achieve China's goals of “carbon peak” and “carbon neutrality.” Flywheel energy storage realizes energy storage through high-speed rotating kinetic energy. Flywheel energy storage offers remarkable advantages, including rapid response, high efficiency, lack of pollution, and extended lifespan.^{1–4} It makes FESS a good candidate for electrical grid regulation to improve distribution efficiency and smoothing power output from renewable energy sources like wind farms.⁵ It is an

internationally recognized high-quality grid frequency modulation resource. In order to enhance energy storage, the flywheel rotor operates at an exceptionally high

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speed, demanding materials with exceptional strength. Due to their high strength, low density, and high modulus, composite materials have become a primary choice for manufacturing flywheel rotors.^{6,7} The optimization design and strength verification calculation of flywheel rotors has become a hot research topic in flywheel energy storage technology.^{8,9}

A lot of research has been carried out on the structural strength of composite materials in various countries. Arnold et al.¹⁰ established an analytical model capable of performing an elastic stress analysis of rotor systems, which considered the pressure surface tractions and body forces (in the form of temperature changes and rotation fields). Ha et al.¹¹ discussed three different rim design cases of a hybrid composite flywheel rotor using strength ratio optimization, and one of these rotors has been successfully manufactured. Wen and Jiang¹² performed a structure optimum design based on the displacement method to maximize the energy storage capacity of a hybrid composite multi-ring flywheel rotor. Conteh and Nsofor⁷ conducted a mechanical analysis study on materials suitable for high-speed rotating flywheels and showed that the rapid development of new composite materials is of great help to flywheel energy storage to improve energy density. Skinner and Mertiny¹³ discussed the effect of viscoelasticity on stress relaxation between the rotor flange and hub of a composite flywheel under different operating conditions. Li¹⁴ designed and discussed different process methods and analytical models of carbon fiber composite flywheel rotor and analyzed the stress distribution, energy storage density optimization, and dynamic characteristics of the flywheel. Chen et al.¹⁵ used the stiffness attenuation model to predict the failure process of composite flywheel under different conditions. Pérez-Aparicio and Ripoll¹⁶ proposed the closed-form expressions of the displacement, stress, and failure factors of the composite flywheel rotor. Kim et al.¹⁷ designed and manufactured a multi-material rim-mounted composite rotor with a speed of up to 15,000 rpm. Tang et al.¹⁸ designed a multi-layer hybrid flywheel and discussed the influence of the number of layers on stress and deformation. Hiroshima et al.¹⁹ discussed the connection between three kinds of rims and hubs of composite flywheels and analyzed the key factors causing vibration. Dai et al.²⁰ proposed a ring plain fabric structure with both circumferential and radial reinforcement. Filippatos et al.²¹ designed a laminated composite flywheel rotor. However, the flywheel rotors discussed in the above articles are relatively small in size and low in energy storage.

In 2011, Beacon Power Company of the United States developed a carbon fiber composite flywheel energy storage product with a storage energy of 25 kWh and a rated power of 100 kW, which has been successfully applied to power system frequency modulation.²²

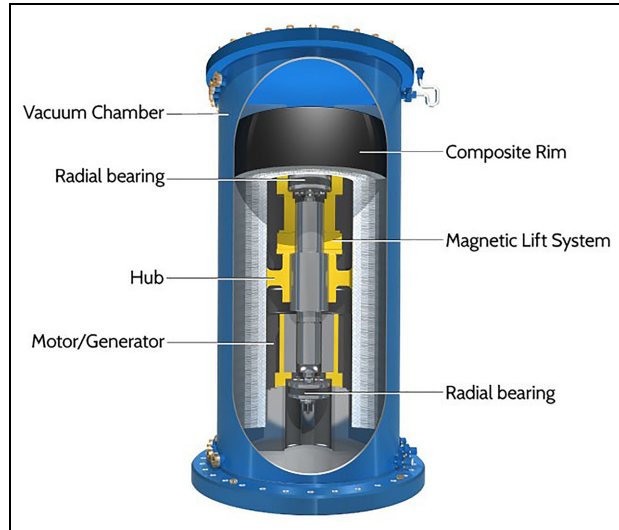


Figure 1. Schematic diagram of Beacon Power's carbon fiber flywheel energy storage unit.

The schematic diagram of the flywheel energy storage unit is shown in Figure 1, and the carbon fiber composite rim is the main energy storage part. However, there are few discussions on the strength of large-size composite flywheel rotors with this energy level in public literature at home and abroad. The composite flywheel energy storage rotor usually adopts a hollow cylinder structure. The outer rim is a composite material, and the interior is connected to the central shaft through a metal material hub to achieve torque transmission. The composite material rim is the main energy storage part. Because the composite material adopts a multi-layer ply structure, it has anisotropic characteristics. The circumferential strength is much higher than the radial strength. In the high-speed rotation state, the composite material rim structure bears a large centrifugal load, and the stress distribution is different from the isotropic metal material. Therefore, it is of great significance to accurately calculate the internal stress distribution of the composite material structure and the maximum stress at the highest working speed, and then realize the optimization design of the strength and structure of the flywheel energy storage rotor for improving the energy storage of the composite flywheel energy storage system and developing a large-capacity flywheel energy storage system suitable for power system applications.

In this paper, the large-size composite rotor of a 100 kW/25 kWh flywheel energy storage system by Beacon Power Company is taken as the object. Based on the basic principle of elastic mechanics, the stress distribution formula of anisotropic flywheel rotor under high-speed rotation is derived. Based on the principle of stress superposition, the stress analysis formula of composite flywheel fixed speed after interference assembly with metal hub is obtained. According to the radial

displacement difference of each ring, the appropriate interference amount is selected, and then the finite element analysis model is established to simulate the stress distribution of the rotor. By comparing with the analytical results, the rationality of the model is verified. The differences between the two-ring isomorphism and the two-ring isomerism are analyzed and compared. It is further verified that the analytical results and the simulation results are applicable to the three-ring and four-ring composite flywheels. The results show that the radial stress distribution of the composite flywheel rotor can be improved by improving the inner and outer diameters and the anisotropy values, setting the isomerism structure and the multi-ring structure. The results show that for the large-size composite flywheel, the existing formula analysis calculation and finite element simulation calculation results are consistent with each other, but there is a lack of experimental verification, which is the future research direction.

Stress distribution of composite flywheel rotor

Basic equations of mechanics of anisotropic elastic body

The layup materials of composite material flywheel rotor are generally anisotropic, which is different from the metal material. It can be seen from Li¹⁴ that the results of the plane stress assumption and the plane strain assumption for the flywheel rotor are not much different. Figure 2 shows the structure and cylindrical coordinate setting of the hollow cylinder composite flywheel rotor, where the z -axis is the symmetry axis, the r -axis is the radial direction, and the θ is the rotation direction (circumferential direction). The size of the flywheel rotor studied in the z direction is larger than that in the other two directions, and the flywheel rotor is subjected to a centrifugal force perpendicular to the z direction in the high-speed rotation state, so it can be regarded as a plane strain problem.¹⁴

A micro-element is taken at the radius r of the rotor structure. The stress on the micro-element is shown in Figure 1 when the rotor rotates at the speed ω . The cylindrical flywheel rotor belongs to the axisymmetric structure. According to the basic theory of elastic mechanics, the equilibrium differential equation of axisymmetric flywheel rotor is¹⁴:

$$\begin{cases} \frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} + \rho r \omega^2 = 0 \\ \frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = 0 \end{cases} \quad (1)$$

In the equation, σ_r , σ_θ , σ_z are the normal stress in the radial, circumferential, and axial directions, respectively;

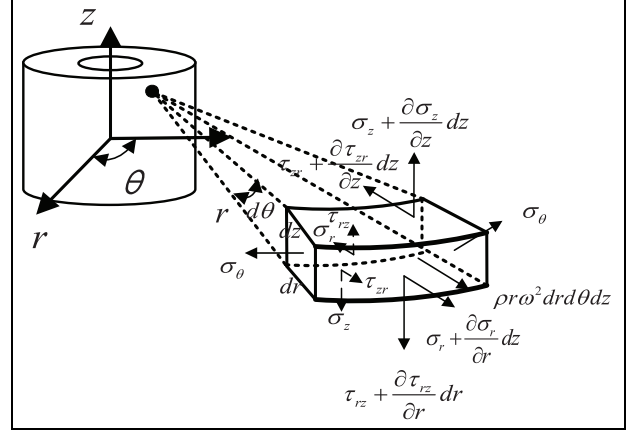


Figure 2. Microelement force diagram in cylindrical coordinate system.

r is the radius, ρ is the density of the composite material, ω is the rotational speed.

According to the relationship between strain and displacement of axisymmetric structure, the geometric equation (strain equation) of the axisymmetric problem in cylindrical coordinates can be derived.¹⁴

$$\begin{aligned} \varepsilon_r &= \frac{\partial u_r}{\partial r}, \varepsilon_\theta = \frac{u_r}{r}, \varepsilon_z = \frac{\partial u_z}{\partial z} \\ u_\theta &= 0, \gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \end{aligned} \quad (2)$$

In the equation, ε_r , ε_θ , ε_z are the normal strain in the radial, circumferential, and axial directions, u_r , u_θ , u_z are the radial displacements in the radial, circumferential, and axial directions, γ_{rz} is the radial-axial shear strain. Furthermore, according to the relationship between strain and stress of axisymmetric structure (generalized Hooke's law), the physical equation (constitutive equation) of the potential element can be obtained. For anisotropic material elements, the physical equation under plane strain theory is¹⁴:

$$\begin{cases} \varepsilon_r = \frac{\sigma_r}{E_r} - \nu_{\theta r} \frac{\sigma_\theta}{E_\theta} - \nu_{zr} \frac{\sigma_z}{E_z} \\ \varepsilon_\theta = \frac{\sigma_\theta}{E_\theta} - \nu_{r\theta} \frac{\sigma_r}{E_r} - \nu_{z\theta} \frac{\sigma_z}{E_z} \\ \varepsilon_z = \frac{\sigma_z}{E_z} - \nu_{rz} \frac{\sigma_r}{E_r} - \nu_{\theta z} \frac{\sigma_\theta}{E_\theta} \\ \gamma_{rz} = \frac{\tau_{rz}}{G_{rz}} \end{cases} \quad (3)$$

In the above equation, E_r , E_θ , E_z are the elastic modulus of the composite material in the radial, circumferential, and axial directions; $\nu_{\theta r}$, $\nu_{r\theta}$, ν_{rz} , ν_{zr} , $\nu_{\theta z}$, $\nu_{z\theta}$ are the Poisson's ratios of the composite material in the circumferential-radial, radial-circumferential, radial-axial, axial-radial, circumferential-axial, axial-circumferential directions.

The outer diameter of the flywheel rotor studied in this article is much smaller than the axial height, so the plane strain assumption is adopted. At this point,

$$\begin{aligned} u_z = 0, \varepsilon_z = \frac{\partial u_z}{\partial z} = 0, \gamma_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} = 0, \frac{\partial \sigma_r}{\partial z} \\ = \frac{\partial \sigma_\theta}{\partial z} = \frac{\partial \sigma_z}{\partial z} = 0 \end{aligned} \quad (4)$$

At this point, Formula (1) becomes:

$$\frac{\partial \sigma_r}{\partial r} + \frac{\sigma_r - \sigma_\theta}{r} + \rho r \omega^2 = 0 \quad (5)$$

At this point, Formula (3) becomes:

$$\begin{cases} \varepsilon_r = \frac{\sigma_r}{E_r} - \nu_{\theta r} \frac{\sigma_\theta}{E_\theta} - \nu_{zr} \frac{\sigma_z}{E_z} \\ \varepsilon_\theta = \frac{\sigma_\theta}{E_\theta} - \nu_{r\theta} \frac{\sigma_r}{E_r} - \nu_{z\theta} \frac{\sigma_z}{E_z} \\ \sigma_z = E_z \cdot \left(\nu_{rz} \frac{\sigma_r}{E_r} + \nu_{\theta z} \frac{\sigma_\theta}{E_\theta} \right) \end{cases} \quad (6)$$

Define intermediate variables to simplify equation solving:

$$\begin{aligned} E_{r1} = \frac{E_r}{1 - \nu_{zr}\nu_{rz}}, E_{\theta1} = \frac{E_\theta}{1 - \nu_{\theta z}\nu_{z\theta}}, \nu_{\theta r1} \\ = \frac{\nu_{\theta r} + \nu_{zr}\nu_{\theta z}}{1 - \nu_{\theta z}\nu_{z\theta}}, \nu_{r\theta1} = \frac{\nu_{r\theta} + \nu_{z\theta}\nu_{rz}}{1 - \nu_{zr}\nu_{rz}} \end{aligned} \quad (7)$$

Then the physical equation Formula (6) can be changed to:

$$\begin{cases} \varepsilon_r = \frac{\sigma_r}{E_{r1}} - \frac{\nu_{r\theta1}}{E_{\theta1}} \sigma_\theta \\ \varepsilon_\theta = \frac{\sigma_\theta}{E_{\theta1}} - \frac{\nu_{\theta r1}}{E_{r1}} \sigma_r \end{cases} \quad (8)$$

At this point, the geometric equation is:

$$\begin{cases} \varepsilon_r = \frac{\partial u_r}{\partial r} \\ \varepsilon_\theta = \frac{u_r}{r} \end{cases} \quad (9)$$

Substituting equations (8) and (9) into equation (5), the equilibrium equation of the radial displacement u_r can be obtained:

$$r^2 \frac{d^2 u_r}{dr^2} + r \frac{du_r}{dr} - \frac{E_{\theta1}}{E_{r1}} u_r = -\frac{\rho \omega^2}{E_{r1}} \left(1 - \frac{\nu_{\theta r1}^2 E_{r1}}{E_{\theta1}} \right) r^3 \quad (10)$$

The stress boundary condition of the flywheel rim is that the radial stress of the inner and outer surfaces of the rim is 0:

$$\sigma_r|_{r=r_0, r_1} = 0 \quad (11)$$

The stress distribution of the composite flywheel rotor can be obtained by solving the differential equations (10) and (11):

$$\begin{cases} \sigma_r = \frac{3 + \nu_{\theta r1}}{9 - K_1^2} \rho r^2 \omega^2 \left[\frac{1 - \lambda^{3+K_1}}{1 - \lambda^{2K_1}} \left(\frac{r_i}{r} \right)^{K_1-1} \right. \\ \quad \left. - \frac{\lambda^{2K_1} - \lambda^{3+K_1}}{1 - \lambda^{2K_1}} \left(\frac{r_i}{r} \right)^{-K_1-1} - \left(\frac{r_i}{r} \right)^2 \right] \\ \sigma_\theta = \frac{3 + \nu_{\theta r1}}{9 - K_1^2} \rho r^2 \omega^2 \\ \quad \left[K_1 \cdot \frac{1 - \lambda^{3+K_1}}{1 - \lambda^{2K_1}} \left(\frac{r_i}{r} \right)^{K_1-1} + K_1 \cdot \frac{\lambda^{2K_1} - \lambda^{3+K_1}}{1 - \lambda^{2K_1}} \left(\frac{r_i}{r} \right)^{-K_1-1} \right. \\ \quad \left. - \frac{K_1^2 + 3\nu_{\theta r1}}{3 + \nu_{\theta r1}} \left(\frac{r_i}{r} \right)^2 \right] \\ \sigma_z = \left(\frac{\nu_{rz}}{E_r} \sigma_r + \frac{\nu_{\theta z}}{E_\theta} \sigma_\theta \right) \cdot E_z \end{cases} \quad (12)$$

Further, the radial displacement can be derived:

$$\begin{aligned} u_r = \frac{\rho \omega^2}{E_r} \cdot \frac{(3 + \nu_{\theta r1})(1 - \nu_{\theta r1}\nu_{r\theta1})}{(9 - K_1^2)(1 - \lambda^{2K_1})} \\ \cdot \left[\frac{(1 - \lambda^{3+K_1})r^{K_1}}{(K + \nu_{\theta r1})r_0^{K_1-3}} - \frac{\lambda^2 - \lambda^{3+K_1}}{(-K + \nu_{\theta r1})r_0^{-K_1-3}} - \frac{r^3(1 - \lambda^{2K_1})}{9E_r - E_\theta} \right] \end{aligned} \quad (13)$$

In the above equation, $K_1 = \sqrt{E_{\theta1}/E_{r1}}$, represents the degree of anisotropy of the material, λ is the ratio of the inner and outer diameters of the composite wheel rim, and r_0 is the outer radius of the composite flywheel rotor.

If $K_1 = 1$, the radial and circumferential elastic modulus of the material are equal, the equations (12) and (13) become the calculation formulas of isotropic materials, which can be used to calculate the stress distribution and radial displacement of the metal hub.

Stress generated by interference fit

In order to improve the energy storage of the flywheel system, it is necessary to ensure that the composite rim of the flywheel rotor has a certain thickness. In order to meet the strength requirements of the composite flywheel rotor and improve the stress distribution in the radial direction, a multi-ring structure of different materials is usually used. Each ring is assembled by interference fit, and the interior is connected with the metal material hub by interference fit. For the flywheel rotor rim composed of this multi-ring composite material, in the working speed range, in addition to meeting the strength conditions, it should also meet the deformation coordination conditions, that is, there is no separation between the inner and outer rings, and there must be contact stress. Therefore, the reasonable selection of the

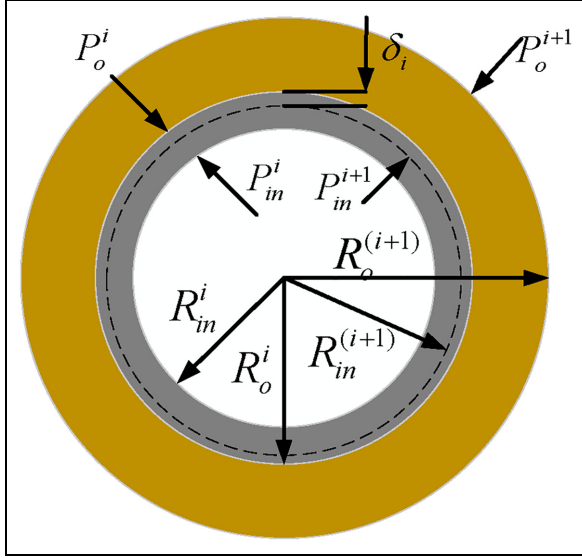


Figure 3. Flywheel rotor interference assembly force diagram.

interference between the inner and outer rings and the rim and the metal hub is very important for the strength design of the flywheel rotor.

The contact force and deformation on the interference fit surface of the inner and outer rings of the rotor composed of multiple rings are shown in Figure 3. In the figure, P_{in}^i and P_o^i are the internal and external contact pressures on the i -th ring respectively, R_{in}^i and R_o^i are the inner and outer radii of the i -th layer respectively, $x_i = r_i/R_o^i$.

For the composite flywheel rotor composed of $i + 1$ rings, when the i -th layer is under the action of external pressure P_i , after the interference assembly with the $(i + 1)$ -th layer, the interference of the initial assembly is:

$$\delta_i = u_r|_{r=R_{in}^{i+1}} - u_r|_{r=R_o^i} \quad (14)$$

Then, the contact external pressure P_i of the i -th layer is:

$$P_i = \delta_i / (c\alpha_i + c\alpha_{i+1}) \quad (15)$$

In the equation:

$$c\alpha_i = \frac{R_o^i \left[(k_i - \nu_{\theta r}^i) + (\lambda_i)^{2k_i} (k_i + \nu_{\theta r}^i) \right]}{E_{\theta}^i \left[1 - (\lambda_i)^{2k_i} \right]}$$

$$c\alpha_{i+1} = \frac{\lambda_{i+1} R_o^{i+1} \left[(\lambda_{i+1})^{2k_{i+1}} (k_{i+1} - \nu_{\theta r}^{i+1}) + (k_{i+1} + \nu_{\theta r}^{i+1}) \right]}{E_{\theta}^{i+1} \left[1 - (\lambda_{i+1})^{2k_{i+1}} \right]} \quad (16)$$

According to the conditions of surface stress and boundary continuity between contact rings under the condition of interference fit, the radial stress, circumferential stress, and radial displacement of the i -th ring rim can be obtained²³:

$$\sigma_r^i = \frac{P_{in}^i (\lambda_i)^{k_i+1} - P_o^i (\lambda_i)^{k_i-1}}{1 - (\lambda_i)^{2k_i}} - \frac{P_{in}^i - P_o^i (\lambda_i)^{k_i-1}}{1 - \lambda_i^{2k_i}} (\lambda_i)^{k_i+1} (x_i)^{-k_i-1}$$

$$\sigma_{\theta}^i = k_i \left[\frac{P_{in}^i (\lambda_i)^{k_i+1} - P_o^i (\lambda_i)^{k_i-1}}{1 - (\lambda_i)^{2k_i}} (x_i)^{k_i-1} + \frac{P_{in}^i (\lambda_i)^{k_i+1} - P_o^i (\lambda_i)^{2k_i}}{1 - (\lambda_i)^{2k_i}} (x_i)^{-k_i-1} \right]$$

$$u_r(x_i, R_i) = \frac{P_{in}^i R_o^i (\lambda_i)^{k_i+1}}{E_{\theta}^i \left[1 - (\lambda_i)^{2k_i} \right]} \left[(k_i - \nu_{\theta r}^i) (x_i)^{k_i} + (k_i + \nu_{\theta r}^i) (x_i)^{-k_i} \right]$$

$$- \frac{P_o^i R_o^i}{E_{\theta}^i \left[1 - (\lambda_i)^{2k_i} \right]} \left[(k_i - \nu_{\theta r}^i) (x_i)^{k_i} + (\alpha_i)^{2k_i} (k_i + \nu_{\theta r}^i) (x_i)^{-k_i} \right] \quad (17)$$

In order to solve the stress distribution generated by the interference fit of the $(i + 1)$ -th layer multi-ring structure, it is necessary to first solve the stress distribution of the first layer and the second layer rim interference assembly, at this time, there is $P_{in}^1 = P_o^2 = 0$, $P_{in}^2 = P_o^1 = P_1$. According to equation (17), the stress distribution σ_r^1 , σ_{θ}^1 and σ_r^2 , σ_{θ}^2 of the flywheel generated by the interference assembly of the first and second rims can be obtained. When solving the first $(i + 1)$ -th layer, only the first i -th layer is regarded as a non-prestressed whole, that is, the influence of the assembly of the first i -th layer is not considered. According to equation (17), the stress distribution of the $(i + 1)$ -th layer and the front i -th layer is solved. Based on the principle of stress superposition, the total stress distribution generated by the interference fit of the $(i + 1)$ -th layer multi-ring structure is obtained by summing the stress distribution of each assembly of the front i layer.

In equation (17), each ring material is suitable for both anisotropic composite materials and isotropic metal materials. For the case of a single composite rim and metal hub, $k_1 = 1$ and $k_2 > 3$ in equation (17) can be used to calculate the stress distribution of the metal hub and single composite rim under interference fit.

Total stress of multi-ring flywheel rotor under constant speed operation

The multi-ring interference fit composite flywheel rotor running at a given speed can calculate the stress distribution inside each ring material based on the principle of stress superposition. The stress distribution of the flywheel rotor under the interference assembly of the composite rim and the metal hub can be obtained by directly adding the stress distribution obtained in equations (12) and (17).

Modeling and simulation of single-ring composite flywheel

Finite element modeling of single-ring composite flywheel rotor

At present, most of the composite flywheel rotor rims on the market are made of carbon fiber/epoxy resin materials. In this section, a composite flywheel rotor composed of a single composite rim and a metal hub is considered. The outer part of the rotor is the rim of the composite material, and the material is CF-T700; the wheel hub matched with the inner edge of the composite rim is made of aluminum alloy 7075 material. The structure size of the flywheel rotor is as follows: the inner diameter of the hub $d_{in} = 300$ mm, the outer diameter of the hub (the inner diameter of the rim) $D_{in} = 460$ mm, the outer diameter of the rim $D_o = 820$ mm, the height of the rotor $H = 1510$ mm, and the maximum working speed of the flywheel rotor is 15,500 rpm. The main performance parameters of CF-T700 composites are shown in Table 1, and the main performance parameters of aluminum alloy are shown in Table 2.

At present, most of the composite flywheel rotor rims on the market are made of carbon fiber/epoxy resin materials. This section considers a composite flywheel rotor composed of a carbon fiber composite rim and a metal hub. The main performance parameters of CF-T700 and CF-T800 composites are shown in Table 1.

At present, the commonly used wheel hub materials include ordinary aluminum alloy, high-strength aluminum alloy, and titanium alloy. Table 2 lists the performance parameters of these materials. In this paper, aluminum alloy 7075 with higher strength and lower density is selected.

The finite element model of the flywheel rotor is established in ANSYS Workbench, in which the composite rim adopts 25 layers of ply structure. The aluminum alloy wheel hub structure adopts a hollow cylinder structure. The finite element model and meshing of the flywheel rotor are shown in Figure 4, in which the composite rotor rim is divided into 63,684 elements. The hub part is divided into 35,625 units; the connection between the outer wall of the hub and the inner wall of the composite rim is treated by friction contact, and the friction coefficient is 0.2.

Single-ring composite flywheel rotor interference selection

In order to study the influence of interference on the contact surface between the flywheel rotor rim and the hub, the contact stress generated by different interference is calculated at 15,500 rpm, as shown in Figure 5. It can be seen from the figure that the radial

Table 1. Composite material performance parameters.^{23–26}

Material properties	CF-T700	CF-T800
Matrix	Epoxy resin	Epoxy resin
Density (kg/m^3)	1590	1600
Circumferential modulus (GPa)	150.7	155
Radial modulus (GPa)	7	9
Axial modulus (GPa)	7	9
Circumferential-radial Poisson's ratio	0.3	0.3
Radius-axial Poisson's ratio	0.36	0.36
Circumferential-axial Poisson's ratio	0.3	0.3
Circumferential-radial shear modulus (GPa)	5.5	5.5
Radial-axial shear modulus (GPa)	4.9	3.4
Circumferential-axial shear modulus (GPa)	5.5	5.5
Circumferential tensile strength (MPa)	3200	2900
Radial tensile strength (MPa)	85	70
Axial tensile strength (MPa)	85	70

Table 2. Performance parameters of several commonly used wheel hub materials.

Performance parameter	Tensile strength (MPa)	Elastic modulus (GPa)	Density (kg/m^3)	Specific strength ($\times 10^6 \text{ m}^2/\text{s}^2$)
Ordinary aluminum alloy	400	160	2800	0.143
Aluminum alloy 7075	524	71	2810	0.213
Titanium alloy	1646	247.6	4620	0.366

compressive stress of the contact surface gradually increases with the increase in the amount of interference. When the amount of interference is less than 0.04 mm, the radial stress of the contact surface is greater than 0, indicating that at a given speed, there is no compressive stress on the contact surface, which does not meet the deformation coordination condition. Therefore, the contact between the contact surfaces cannot be guaranteed, so the amount of interference must be greater than 0.04 mm. However, because the radial compressive stress increases with the increase of the amount of interference, if the amount of interference is too large, the compressive stress of the contact surface may exceed the radial strength of the composite material, which will cause damage to the flywheel.

Since the application of the amount of interference will produce the interface pressure P_i in equation (15), the value can also reflect whether the interface is separated. The interface pressure generated by different interference amounts at 15,500 rpm is calculated, as shown in Figure 5. It can be seen from the diagram that the interface pressure gradually increases with the increase of the amount of interference. When the amount of interference is less than 0.04 mm, the interface pressure is less

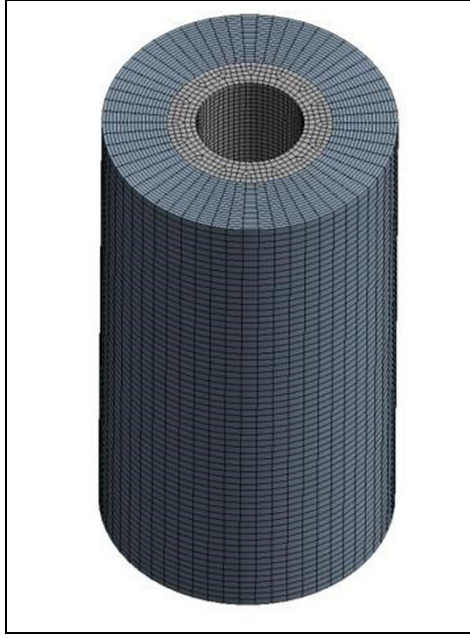


Figure 4. Composite flywheel rotor modeling.

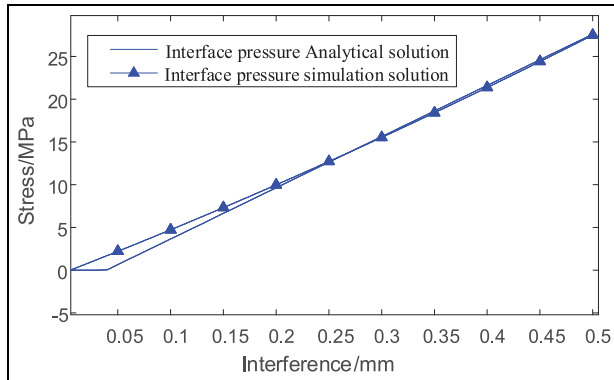


Figure 5. Effect of interference on interface compressive stress.

than or equal to 0, indicating that at a given speed, there is no interface pressure on the contact surface, and the deformation coordination condition is not satisfied. Therefore, the contact between the contact surfaces cannot be guaranteed, so the amount of interference must be greater than 0.04 mm. However, due to the increase of the interface pressure with the increase of the amount of interference, the value of interference should not be too large, and the appropriate range should be selected.

When the flywheel rotor rotates at high speed, the hub and rim will undergo radial deformation. The higher the speed, the greater the deformation. Due to the different materials of the rim and hub, the radial deformation is also different. If the radial deformation of the composite rim is greater than the sum of the radial deformation of the hub and the amount of interference, the hub and the rim will be detached, resulting

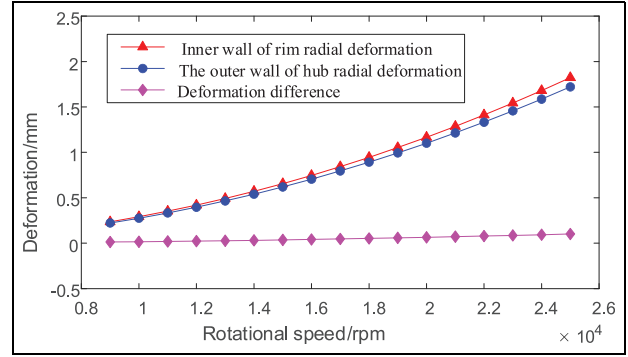


Figure 6. The radial deformation and interference of the flywheel change with the speed.

in a flywheel accident. According to the analytical formula of the radial displacement u_r of the flywheel rotor in equation (13), the analytical solution of the radial deformation of the inner wall of the rim and the outer wall of the hub can be obtained. Figure 6 shows the radial deformation of the contact surface of the flywheel rim and hub at different speeds and the interference of the actual action at this speed.

From the above two diagrams, it can be concluded that the radial deformation of the flywheel increases with the increase of the rotational speed. On the contact surface, due to the different materials, the deformation of the inner edge of the flywheel rim is always greater than the deformation of the outer wall of the flywheel hub, and the difference between the two is also increasing, so that the actual interference between the flywheel rim and the hub is continuously reduced, and the reduction of the interference will lead to the continuous reduction of the radial compressive stress superimposed by the interference assembly. The higher the rotational speed, the faster the trend of the stress reduction. Due to the viscoelasticity of the composite material, the stress relaxation phenomenon may occur in the radial direction of the composite flywheel,²⁶ so that the radial stress caused by the interference fit between the rim and the hub is reduced, which leads to the accident of the flywheel. Therefore, in order to avoid the separation of the rim and hub of the flywheel rotor at high speed, the amount of interference between the composite rim and the metal hub needs to be more redundant than the minimum interference condition. Therefore, the interference amount can be selected as 0.34 mm, which can provide 0.3 mm actual interference at the speed of 15,500 rpm.

Rotor rim stress analysis of single-ring flywheel at constant speed

Firstly, the case of the only composite rim and no metal hub is discussed. A speed of 15,500 rpm around

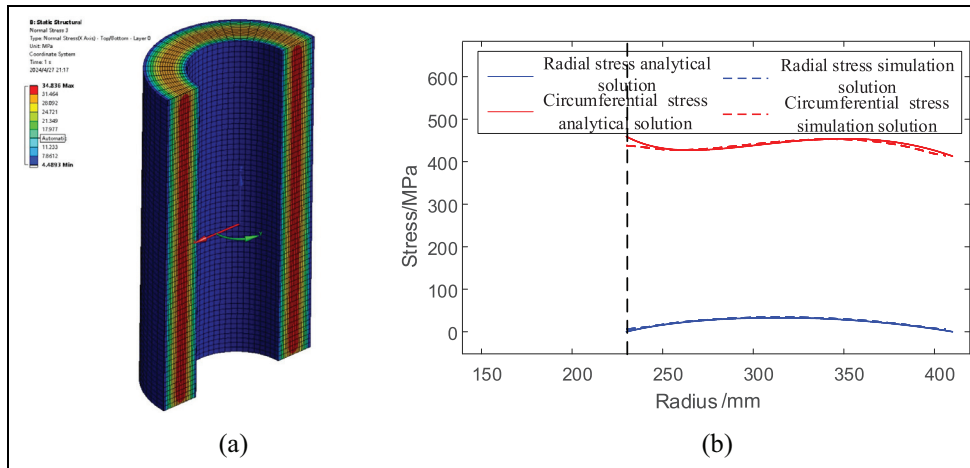


Figure 7. Stress distribution of flywheel rotor rim at 15,000 rpm/min fixed speed: (a) radial stress distribution nephogram and (b) stress distribution curve along the radius.

the z -axis is applied to the flywheel rotor rim to constrain the movement of the inner surface of the hub in the r and θ directions. The stress distribution of the rotor under centrifugal force is calculated, as shown in Figure 7(a). The curves of radial stress and hoop stress varying with the radius of the rotor at the middle height are obtained, as shown in Figure 7(b). The corresponding stresses calculated by the analytical equation (17) are also shown in the figure.

Comparing the finite element simulation results with the analytical calculation results, it can be seen that the error between the two is very small, indicating that the finite element model and the analytical analysis results are very consistent and can be mutually verified. The simulation results show that the circumferential stress and radial stress inside the anisotropic flywheel rotor are very different, and the maximum circumferential stress reaches 459.195 MPa, which occurs on the inner wall of the rim. The maximum radial stress is only 34.28 MPa, which occurs near the middle of the rim (radius 313.3 mm). Because the radial strength of the composite material is much lower than the hoop strength, although the maximum radial stress is low, it is easier to exceed the radial tensile strength limit, resulting in structural damage. According to the strength of the composite CF-T700 given in Table 1, at the maximum working speed of 15,500 rpm, the circumferential safety factor is 6.97, while the radial safety factor is only 2.48. The safety factors in the two directions are quite different, so it is necessary to pay special attention to the stress in the radial direction.

Total stress distribution of single-ring flywheel rotor

Considering the interference fit between the composite rim and the aluminum alloy wheel hub, the interference

fit of 0.34 mm is applied to the mating surface in the model. At the speed of 15,500 rpm, the stress distribution of the composite rotor rim and the metal wheel hub is calculated, as shown in Figure 8(a). The curves of radial stress and circumferential stress at the middle height with the radius of the rotor are extracted, as shown in Figure 8(b). The corresponding stresses calculated according to the superposition of analytical equations (12) and (17) are also shown in the figure. It can be seen that under the condition of considering the interference fit, the difference between the finite element analysis results and the analytical calculation results is still very small, which can be verified by each other well.

From the calculation results of stress distribution, it can be seen that:

- (1) The radial stress on the contact surface between the flywheel rim and the hub is less than 0, which indicates that there is still compressive stress on the contact surface between the hub and the rim at 15,500 rpm. The contact surface satisfies the deformation coordination condition and will not be separated. The maximum radial stress still occurs near the middle position of the rim (radius 322 mm), and the maximum value is about 31.9 MPa, which is slightly smaller than the maximum radial stress value of the pure composite rim, indicating that the interference fit can reduce the maximum radial stress in the composite rim.
- (2) On the interference fit surface, the circumferential stress has a sudden change. The circumferential stress in the hub is small, while the circumferential stress of the composite rim increases sharply. The maximum circumferential

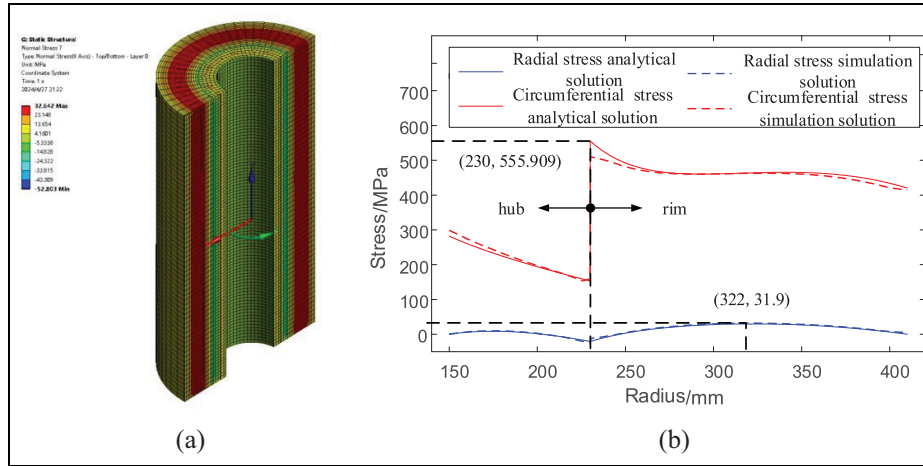


Figure 8. Stress distribution of flywheel rotor hub and rim interference fit at constant speed: (a) radial stress distribution nephogram and (b) stress distribution curve along the radius.

stress still occurs on the inner wall of the rim, reaching 555.909 MPa, which is higher than the maximum circumferential stress of the pure composite rim. It shows that the interference fit increases the maximum circumferential stress in the composite rim.

Modeling and simulation of the two-ring composite flywheel rotor

At present, in order to solve the problem of insufficient radial strength, most of the flywheel rotors with high energy storage density adopt a multi-ring composite structure to reduce the radial thickness of a single ring, aiming to achieve a balanced distribution of the overall stress under high-speed rotation. The material and structure of each component ring directly affect the improvement of energy storage. Therefore, it is necessary to study the stress distribution of the multi-ring composite flywheel rotor. If the material of the multi-ring rotor is the same, it is called multi-ring isomorphism; if the material of the multi-ring rotor is not the same, it is called multi-ring isomerism.

Finite element modeling of a two-ring composite flywheel rotor

The finite element model of the flywheel rotor is established in ANSYS Workbench. The outer of the rotor is a composite material wound rim. The material of the ring 1 is CF-T700, and the material of the ring 2 is CF-T700 or CF-T800. The thickness distribution of each ring is equal. The wheel hub, matched with the inner edge of the composite rim, is made of aluminum alloy 7075 material. The structure size of the flywheel rotor is as follows: the inner diameter of the hub $d_{in} = 300$ mm,

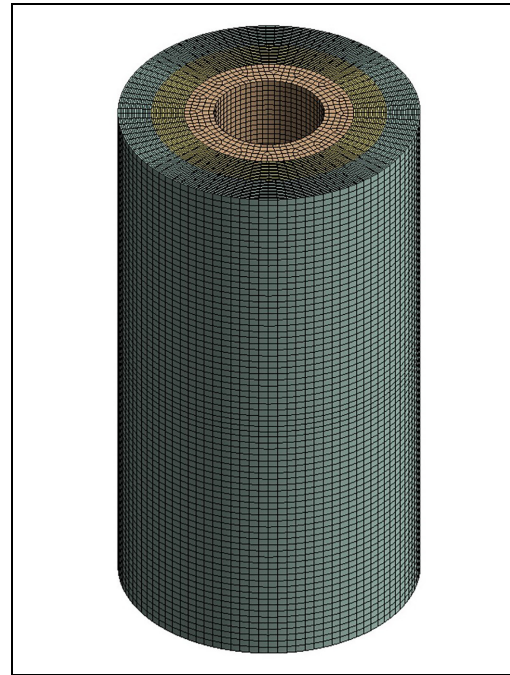


Figure 9. Two-ring composite flywheel rotor modeling.

the outer diameter of the hub (the inner diameter of the ring 1 rim) $d_o = d_{in1} = 460$ mm, the inner diameter of the ring 2 rim (the outer diameter of the ring 1 rim) $D_{o1} = d_{in2} = 640$ mm, the outer diameter of the ring 2 rim $D_{o2} = 820$ mm, the rotor height $H = 1510$ mm, and the maximum working speed of the flywheel rotor is 15,500 rpm.

Among them, the two rings of the composite wheel rim adopt an 18-layer layer structure respectively; the aluminum alloy wheel hub structure adopts a hollow cylinder structure. The finite element model and mesh generation of the flywheel rotor are shown in Figure 9,

Table 3. Displacement and interference of each ring of a two-ring composite flywheel.

Location		Inner surface displacement (mm)	Outer surface displacement (mm)	Displacement difference at the interface (mm)	The amount of interference (mm)
Isomorphism analysis	Hub	0.748	\	\	\
	Ring 1	0.555	0.107	0.2	0.2
	Ring 2	1.344	-0.748	1.05	1.05
Isomorphism simulation	Hub	0.754	\	\	\
	Ring 1	0.557	0.11	0.2	0.2
	Ring 2	1.350	-0.753	1.05	1.05
Isomerism simulation	Hub	0.748	\	\	\
	Ring 1	0.555	0.107	0.2	0.2
	Ring 2	1.344	-0.748	1.05	1.05
Isomerism simulation	Hub	0.748	\	\	\
	Ring 1	0.555	0.107	0.2	0.2
	Ring 2	1.344	-0.748	1.05	1.05

in which the rim part of the composite rotor ring 1 is divided into 98,496 elements; the rim part of ring 2 is divided into 138,168 units. The hub part is divided into 35,625 units; the connection between the outer wall of the hub and the inner wall of the composite ring 1 rim, the outer wall of the composite ring 1 rim, and the inner wall of the ring 2 rim are all treated by friction contact, and the friction coefficient is 0.2.

Interference selection of two-ring composite flywheel rotor

When the flywheel rotates at high speed, the displacement of the outer diameter of the hub and the displacement of the inner and outer walls of the composite rotor ring 1 and ring 2 rims under the action of centrifugal force are obtained respectively. When the displacement difference at the interface reaches the initial interference (the radial stress of the interface is 0 at this time), radial detachment occurs.

Therefore, in order to prevent the separation between the mounting shaft and the composite rotor at high speed, the minimum interference in the static state must be greater than the difference between the displacement of the inner wall of the outer ring and the displacement of the outer wall of the inner ring. The displacement values of the outer wall of the hub and the inner and outer surface of each ring of the two-ring isomorphism and isomerism composite rotor hub are shown in Table 3.

In order to ensure that the actual interference is 0.3 mm at the maximum working speed of 15,500 rpm, according to the displacement difference of the contact surface between the hub and the ring 1 rim, the ring 1 rim and the ring 2 rim obtained by calculation and simulation, the interference during assembly is set to 0.2 and 1.05 mm respectively.

Finite element simulation of two-ring composite flywheel

Considering the interference fit between the multi-ring isomorphism composite rim and the aluminum alloy hub, the interference fit shown in Table 3 is applied to the mating surface in the model. At the speed of 15,500 rpm, the stress distribution of the composite rotor rim and the metal hub is calculated as shown in Figure 10(a). The variation curves of radial stress and hoop stress with rotor radius at the middle height are extracted, as shown in Figure 10(b), and the corresponding stresses calculated by superposition of analytical equations (12) and (17) are also shown in the diagram. It can be seen that under the condition of considering the interference fit, the difference between the finite element analysis results and the analytical calculation results is still very small, which can be verified by each other well.

Different from the two-ring isomorphism, the two-ring isomerism ring 2 material is replaced with CF-T800 and the above calculation process is repeated to obtain the stress distribution of the two-ring isomerism composite rotor rim and metal hub as shown in Figure 11(a). The variation curves of radial stress and hoop stress with rotor radius at the middle height are extracted, as shown in Figure 11(b). The corresponding stresses calculated by the superposition of analytical equations (12) and (17) are also shown in the figure. It can also be seen that under the condition of considering the interference fit, the difference between the finite element analysis results and the analytical calculation results is still very small, and they can be verified by each other well.

The results show that:

- (1) The radial stress on the contact surface of the flywheel ring 1 rim and hub, ring 1 rim and ring

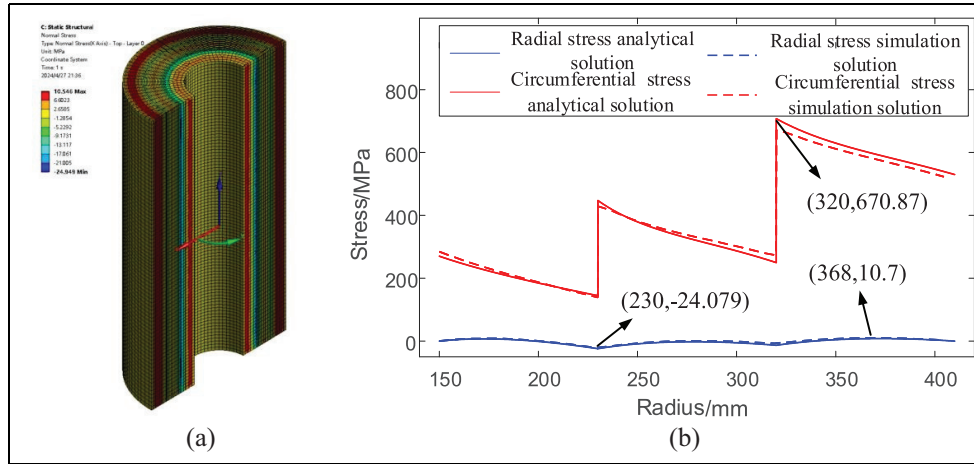


Figure 10. Stress distribution of two-ring isomorphism composite flywheel rotor: (a) radial stress distribution nephogram and (b) stress distribution curve along the radius.

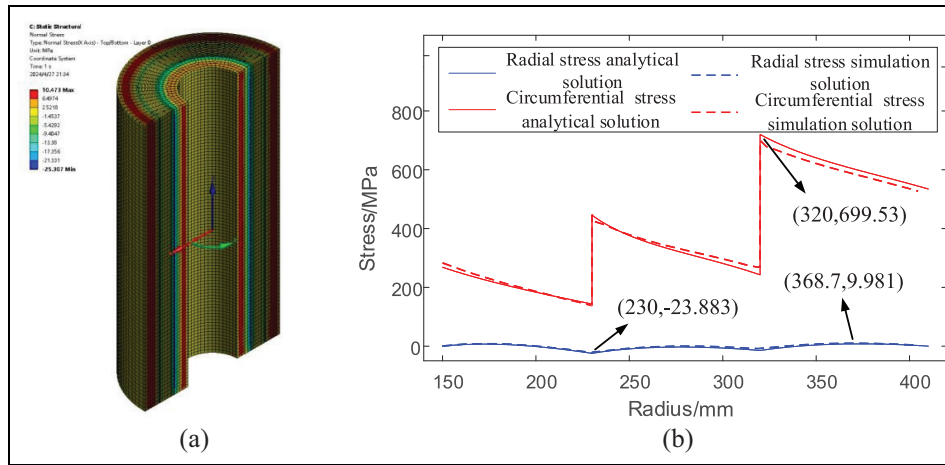


Figure 11. Stress distribution of two-ring isomerism composite flywheel rotor: (a) radial stress distribution nephogram and (b) stress distribution curve along the radius.

2 rim is less than 0, which indicates that there is still compressive stress on the contact surface of the hub and rim at 15,500 rpm. The contact surface meets the deformation coordination condition and will not be separated. The maximum radial stress of the isomorphism structure still occurs near the middle of the ring 2 rim (radius 368 mm), and the maximum value is about 10.7 MPa, which is less than the maximum radial stress of the pure single-ring composite rim. The maximum radial stress of the isomerism structure still occurs near the middle position of the ring 2 rim (radius 368.7 mm), and the maximum value is about 9.98 MPa, which is less than the maximum radial stress value of the multi-ring isomorphism composite material rim, indicating that the multi-ring interference fit can reduce the maximum radial stress in the composite material rim.

- (2) On the interference fit surface, the circumferential stress has a sudden change. The circumferential stress in the hub is small, while the circumferential stress of the composite rim increases sharply. The maximum circumferential stress still occurs on the inner wall of each rim, and the isomorphic structure reaches 670.87 MPa, which is higher than the maximum circumferential stress of the single-ring composite rim. The isomerism structure reaches 699.53 MPa, indicating that the multi-ring interference fit increases the maximum circumferential stress in the composite rim.

Comparison of single-ring, two-ring isomorphism, and two-ring isomerism results

In order to intuitively express the improvement of stress distribution directly by single ring, two ring isomorphism, and two ring isomerism, the maximum value and

Table 4. Comparison of single-ring, multi-ring isomorphism, and multi-ring isomerism results.

Content	Single-ring	Two-ring isomorphism	Two-ring isomerism
Position of maximum circumferential stress (mm)	230	320	320
Maximum circumferential stress (MPa)	555.909	670.87	699.53
Position of maximum radial compressive stress (mm)	230	230	230
Maximum radial compressive stress (MPa)	−18.306	−23.446	−23.883
Position of maximum radial tensile stress (mm)	320	368	368.7
Maximal radial stress (MPa)	31.9	10.07	9.981
Radial safety factor	2.664	7.944	7.013
Circumferential safety factor	5.756	4.77	4.146

maximum position of their radial stress and circumferential stress are listed in the table, as shown in Table 4:

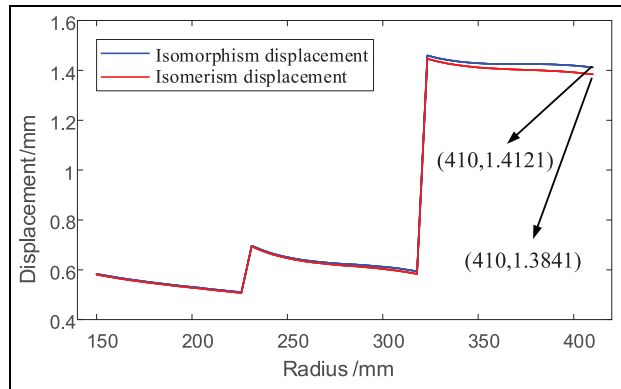
From the results of the above table, it can be seen that:

- (1) Compared with the single-ring and two-ring, the multi-ring structure can greatly improve the radial safety factor under the condition that the circumferential safety factor is slightly reduced. Due to the low radial strength of the composite flywheel, the advantages of the multi-ring structure improve the radial stress distribution of the flywheel;
- (2) Comparing the two-ring isomorphism and the two-ring isomerism, the increase of the elastic modulus of the outer ring can slightly increase the maximum circumferential stress, and the increase of the elastic modulus of the outer ring can slightly decrease the maximum radial stress. Therefore, the increase of the elastic modulus of the outer ring can not only reduce the difficulty of interference assembly but also improve the radial stress distribution of the composite flywheel rotor with interference fit.

Multi-ring isomorphism and multi-ring isomerism rotor radial displacement

In order to maintain the sealing of the rotor, the outer shell of the rotor is protected and sealed. Because the rotor is in a high-speed rotating state, the outermost edge of the rotor will deform outward. In order to avoid the friction between the outermost edge of the rotor and the rotor shell, the curves of the displacement of the multi-ring homogeneous and multi-ring isomerism composite rotor along the radius are extracted, as shown in Figure 12.

From the analytical solution of radial displacement at constant speed and the analytical solution of radial displacement of assembly deformation, it can be obtained that the outermost ring displacement is 1.374 mm when the multi-ring rotor material is the same, and the outermost ring displacement is 1.340 mm

**Figure 12.** Curves of displacement of multi-ring isomorphic and multi-ring isomerism composite rotor along the radius.

when the material is different. It can be seen from the above figure that when the outer ring material is replaced with a material with a large elastic modulus, the deformation of the outer ring of the rotor can be improved, and the friction between the rim and the shell can be reduced. Therefore, even if the tilt of the rotor is not considered, a suitable gap must be left directly between the rotor and the shell.

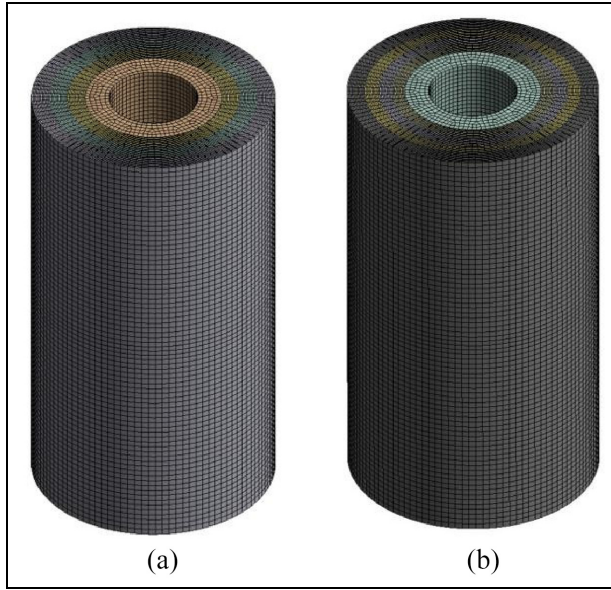
Stress analysis of three-ring and four-ring composite flywheel rotor

Three-ring and four-ring composite flywheel rotor modeling and interference selection

In order to further verify the correctness of the model, the simulation modeling of the three-ring and four-ring interference composite flywheel rotor is carried out. The outer surface of the three-ring rotor is a composite material wound rim. CF-T700 is selected as the material of ring 1, ring 2, and ring 3, and the thickness distribution of each ring rim is equal. The wheel hub, matched with the inner edge of the composite rim, is made of aluminum alloy 7075 material. The structure size of the flywheel rotor is: hub inner diameter $d_{in} = 300$ mm, hub outer diameter (ring 1 rim inner diameter) $d_o = D_{in1} = 460$ mm,

Table 5. Displacement and interference of each ring of multi-ring composite flywheel.

Location		Inner surface displacement (mm)	Outer surface displacement (mm)	Displacement difference at the interface (mm)	The amount of interference (mm)
Three-ring	Hub	0.748	0.662	\	\
	Rim1	0.555	0.486	0.174	0.13
	Rim2	1.344	0.900	−0.432	0.73
	Rim3	0.754	1.503	−0.642	0.94
Four-ring	Hub	0.557	0.667	\	\
	Rim1	1.350	0.442	0.206	0.09
	Rim2	0.748	0.721	−0.298	0.60
	Rim3	0.555	1.100	−0.409	0.71
	Rim4	1.344	1.594	−0.537	0.84

**Figure 13.** Modeling of multi-ring composite flywheel rotor: (a) three-ring composite rotor and (b) four-ring composite rotor.

ring 2 rim inner diameter (ring 1 rim outer diameter) $D_{o1} = D_{in2} = 580$ mm, ring 3 rim inner diameter (ring 2 rim outer diameter) $D_{o2} = D_{in3} = 700$ mm, ring 3 rim outer diameter $D_{o3} = 820$ mm, rotor height $H = 1510$ mm, the maximum working speed of the flywheel rotor is 15,500 rpm.

The four-ring composite flywheel rotor material is the same as the above three rings, and the rim is also distributed with equal thickness. The structure size of the flywheel rotor is: hub inner diameter $d_{in} = 300$ mm, hub outer diameter (ring 1 rim inner diameter) $d_o = D_{in1} = 460$ mm, ring 2 rim inner diameter (ring 1 rim outer diameter) $D_{o1} = D_{in2} = 550$ mm, ring 3 rim inner diameter (ring 2 rim outer diameter) $D_{o2} = D_{in3} = 640$ mm, ring 4 rim inner diameter (ring 3 rim outer diameter) $D_{o3} = D_{in4} = 730$ mm, ring 4 rim outer diameter $D_{o4} = 820$ mm, rotor height $H = 1510$ mm, the maximum working speed of the flywheel rotor is 15,500 rpm.

According to the principle of interference selection in 3.2, the displacement and interference of three-ring and four-ring composite flywheel rotors are analyzed (Table 5).

The finite element model of the flywheel rotor is established in ANSYS Workbench, in which the three rings of composite rim adopt 12 layers of ply structure. Each of the four rings of the composite rim adopts a nine-layer layer structure; the aluminum alloy wheel hub structure adopts a hollow cylinder structure. The remaining settings are the same as in 2.1 above. The finite element model and mesh generation are shown in Figure 13.

Total stress distribution of three-ring and four-ring composite flywheel rotor

Considering the interference fit between the multi-ring composite rim and the aluminum alloy wheel hub, the interference fit shown in Table 4 is applied to the fitting surface in the model. At the speed of 15,500 rpm, the stress distribution of the composite rotor rim and the metal wheel hub is calculated. The curves of radial stress and circumferential stress at the middle height with the radius of the rotor are extracted, as shown in Figure 14. The corresponding stresses calculated by the superposition of analytical equations (12) and (17) are also shown in the figure. It can be seen that under the condition of considering the interference fit, the finite element analysis results of the multi-ring composite flywheel are still very different from the analytical calculation results, which can be verified by each other well.

Comparing the stress distribution calculation results with the stress distribution calculation results of single-ring and two-ring composite flywheel rotors, it can be seen that:

- (1) Under the CF-T700 material, the maximum circumferential stresses of the three-ring and four-ring flywheel rotors are 748.98 and 764.17 MPa. And the maximum stress position

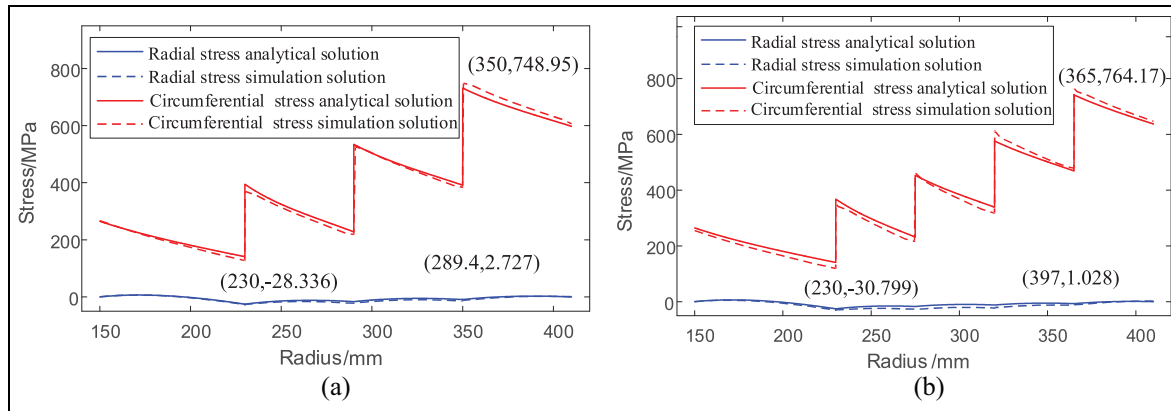


Figure 14. Distribution of stress with a radius of multi-ring composite flywheel rotor: (a) three-ring composite rotor and (b) four-ring composite rotor.

always appears at the inner wall position of the outermost ring. The maximum radial stresses of the three-ring and four-ring flywheel rotors are 2.727 and 1.028 MPa. The radial maximum tensile stress decreases with the increase of the number of rings. It can be predicted that after the number of rings reaches a certain degree, there is only radial compressive stress at 15,500 rpm. When the number of rings reaches a certain degree, the radial compressive stress can not be ignored, and it must be considered whether the radial compressive stress reaches the stress limit.

- (2) Through calculation, the circumferential safety factor of the three-ring composite flywheel rotor is 4.273, and the radial safety factor is 31.17. The circumferential safety factor of the four-ring rotor is 4.188, the radial safety factor is 82.685, and the structure is unreasonable. Therefore, it is necessary to optimize and reconsider the selection of the interference amount.
- (3) Considering the shortcomings of the multi-ring composite flywheel: First, the composite rim needs to be assembled many times, and the deformation of the metal hub and the rim is difficult to coordinate; second, the initial stress of interference assembly will make the resin matrix prone to creep due to its large viscoelasticity; third, the assembly process is difficult, so the number of rings cannot be increased without limit.²⁴

Conclusion

In this paper, the composite flywheel rotor is taken as the object, and the main contents are summarized as follows:

- (1) There is a significant difference in radial and circumferential stresses within the composite rotor flange structure, with radial stresses much lower than circumferential stresses. However, due to the anisotropic properties of composite materials, under certain structural size conditions, the safety factor for radial and circumferential strength is different. Therefore, it is recommended to use radial and circumferential strength equivalent structural designs as much as possible. For multi-ring heterogeneous rotors, increasing the elastic modulus of the outer ring not only reduces the difficulty of interference assembly, but also improves the radial stress distribution of the composite flywheel rotor with an interference fit.
- (2) The rotor speed has different effects on the radial deformation of the composite rim and the metal hub, which in turn affects the contact state of the mating surface. The rotation of the interference fit surface is too small, which may lead to the separation of the composite wheel rim and the metal wheel hub at high speed. If the interference of the mating surface is too large, the local stress of the contact surface composite material may exceed the standard, resulting in material damage. Therefore, a reasonable amount of interference must be selected. For the filament-wound composite flywheel, its lower radial strength greatly restricts the radial thickness. In order to solve the problem of insufficient radial strength of the filament-wound composite flywheel, interference assembly can be used between the rings to generate interlayer pre-compression stress and suppress the radial tensile stress generated by the flywheel rotation.

- (3) To solve the problem of insufficient radial strength, multi-ring combination structures are mostly used to reduce the radial thickness of a single ring in order to achieve a balanced distribution of overall stress under high-speed rotation. However, under the same size, the multi-ring interference fit increases the maximum circumferential stress inside the composite material wheel flange. At the same time, due to the coordination of the radial deformation of the multi-ring rotor with the deformation of the flywheel hub, the calculation becomes complex when there are too many rings. At this time, it is not only easy to cause creep but also greatly increases the process difficulty during assembly. Therefore, it is necessary to increase the number of rings without limitation. In general, it is reasonable to design a composite flywheel rotor with three or four rings.

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References

1. Bolund B, Bernhoff H and Leijon M. Flywheel energy and power storage systems. *Renew Sustain Energy Rev* 2007; 11: 235–258.
2. Dai X, Wei K, Zhang X, et al. A review on flywheel energy storage technology in fifty years. *Energy Storage Sci Technol* 2018; 7: 765–782.
3. Mousavi GSM, Faraji F, Majazi A, et al. A comprehensive review of Flywheel Energy Storage System technology. *Renew Sustain Energy Rev* 2017; 67: 477–490.
4. Dragoni E. Mechanical design of flywheels for energy storage: a review with state-of-the-art developments. *Proc IMechE, Part L: J Materials: Design and Applications* 2019; 233: 995–1004.
5. Li X and Palazzolo A. A review of flywheel energy storage systems: state of the art and opportunities. *J Energy Storage* 2022; 46: 103576.
6. Dai X and Meng Y. Research progress on structural strength technology of composite energy storage flywheel. *Mech Eng* 2005; 4: 7–9.
7. Conteh MA and Nsofor EC. Composite flywheel material design for high-speed energy storage. *J Appl Res Technol* 2016; 14: 184–190.
8. Jiang L, Zhang W and Ma GJ. Shape optimization of energy storage flywheel rotor. *Struct Multidiscipl Optim* 2017; 55: 739–750.
9. Kale V, Thomas M and Secanell M. On determining the optimal shape, speed, and size of metal flywheel rotors with maximum kinetic energy. *Struct Multidiscipl Optim* 2021; 64: 1481–1499.
10. Arnold SM, Saleeb AF and Al-Zoubi NR. Deformation and life analysis of composite flywheel disk systems. *Compos B Eng* 2002; 33: 433–459.
11. Ha SK, Kim SJ and Nasir SU. Design optimization and fabrication of a hybrid composite flywheel rotor. *Compos Struct* 2012; 94: 3290–3299.
12. Wen S and Jiang S. Optimum design of hybrid composite multi-ring flywheel rotor based on displacement method. *Compos Sci Technol* 2012; 72: 982–988.
13. Skinner M and Mertiny P. Effects of viscoelasticity on the stress evolution over the lifetime of filament-wound composite flywheel rotors for energy storage. *Appl Sci* 2021; 11: 9544.
14. Li S. Study on mechanical properties of carbon fiber composite high-speed rotor and optimization of energy storage density. *Changchun Institute of Optics, Fine Mechanics and Physics*, 2003.
15. Chen Q, Li C and Tie Y. Progressive failure analysis of composite flywheel rotor based on progressive damage theory. *J Mech Eng* 2013; 12: 60–65.
16. Pérez-Aparicio JL and Ripoll L. Exact, integrated and complete solutions for composite flywheels. *Compos Struct* 2011; 5: 1404–1415.
17. Kim SJ, Hayat K and Nasir SU. Design and fabrication of hybrid composite hubs for a multi-rim flywheel energy storage system. *Compos Struct* 2014; 107: 19–29.
18. Tang C, Dai X and Wang Y. Mechanical design and rotation test of multilayer hybrid composite flywheel. *J Tsinghua Univ (Nat Sci Ed)* 2015; 3: 361–367.
19. Hiroshima N, Hatta H, Koyama M, et al. Spin test of three-dimensional composite rotor for flywheel energy storage system. *Compos Struct* 2016; 136: 626–634.
20. Dai X, Wei K and Wang Y. Prediction and measurement of elastic parameters of plain woven laminated composite flywheel. *J Compos Mater* 2019; 12: 2833–2842.
21. Filippatos A, Grüber B and Lich J. Design and testing of polar-orthotropic multi-layered composites under rotational load. *Mater Des* 2021; 207: 109853.
22. Ruddell AJ. Flywheel energy storage technologies for wind energy systems. In: Kaldellis JK (ed.) *Stand-alone and hybrid wind energy systems: technology, energy storage and applications*. Cambridge, UK: Woodhead Publishing Limited, 2010, pp.366–392.

23. Yang L and Ren Z. Stress analysis and optimal design of multi-layer composite flywheel. *Mech Sci Technol Aerosp Eng* 2021; 40: 1863–1870.
24. Dai X, Jiang X and Zhang K. *Flywheel energy storage system technology and engineering application*. Beijing: Chemical Industry Press, 2021.
25. Chen Q. *Failure analysis and strength prediction of composite flywheel rotors*. Zhengzhou University, 2014.
26. Hui P, Zu L and Li S. Research on high-speed flywheel rotor with high tension winding carbon fiber composite material. *Comp Sci Eng* 2018; 3: 5–12.

Appendix

Notation

$\sigma_r, \sigma_\theta, \sigma_z$	the normal stress in the radial, circumferential, and axial directions
r	the radius
ρ	the density of the composite material
ω	the rotational speed
$\varepsilon_r, \varepsilon_\theta, \varepsilon_z$	the normal strain in the radial, circumferential, and axial directions
u_r, u_θ, u_z	the radial displacements in the radial, circumferential, and axial directions
γ_{rz}	the radial-axial shear strain
E_r, E_θ, E_z	the elastic modulus of the composite material in the radial, circumferential, and axial directions
$\nu_{\theta r}, \nu_{r\theta}, \nu_{rz}, \nu_{zr}, \nu_{\theta z}, \nu_{z\theta}$	the Poisson's ratios of the composite material in the circumferential-radial, radial-circumferential, radial-axial, axial-radial, circumferential-axial, axial-circumferential directions
λ	the ratio of the inner and outer diameters of the composite wheel rim
r_0	the outer radius of the composite flywheel rotor
r_i	the inner radius of the composite flywheel rotor
p_{in}^i, p_o^i	the internal and external contact pressures on the i -th ring
R_{in}^i, R_o^i	the inner and outer radii of the i -th ring
p_i	the contact external pressure
$\sigma_r^i, \sigma_\theta^i$	the stress of the i -th ring generated by the interference assembly
